

ENGEN370 Environmental Impact Assessment (One)

Summary

- Challenges We Face and Why We Need an EIA process
- Environmental Impact Assessment
 - Nature of an EIA
 - Purpose of an EIA
 - The EIA process
- Environmental Management
 - ISO14001 family
- Multi choice quizzes and report

Challenges We Face

- Natural resources and sustainable management
- 21st century problems
- Population growth and
 - Deforestation
 - Food security
 - Water/Waste
 - Urbanisation
 - Poverty
- Environmental degradation

Natural Resources & Sustainable Management

Managing the use, development, and protection of natural and physical resources in a way, or at a rate, which enables people and communities to provide for their social, economic, and cultural well being and for their health and safety while:

- a) sustaining the potential of natural and physical resources to meet the reasonably foreseeable needs of future generations.
- b) safeguarding the life-supporting capacity of air, water, soil, and ecosystems.
- c) avoiding, mitigate, or remedy any adverse effects of activities on the environment.

Resource Management Act (RMA)1991

21st Century Problems

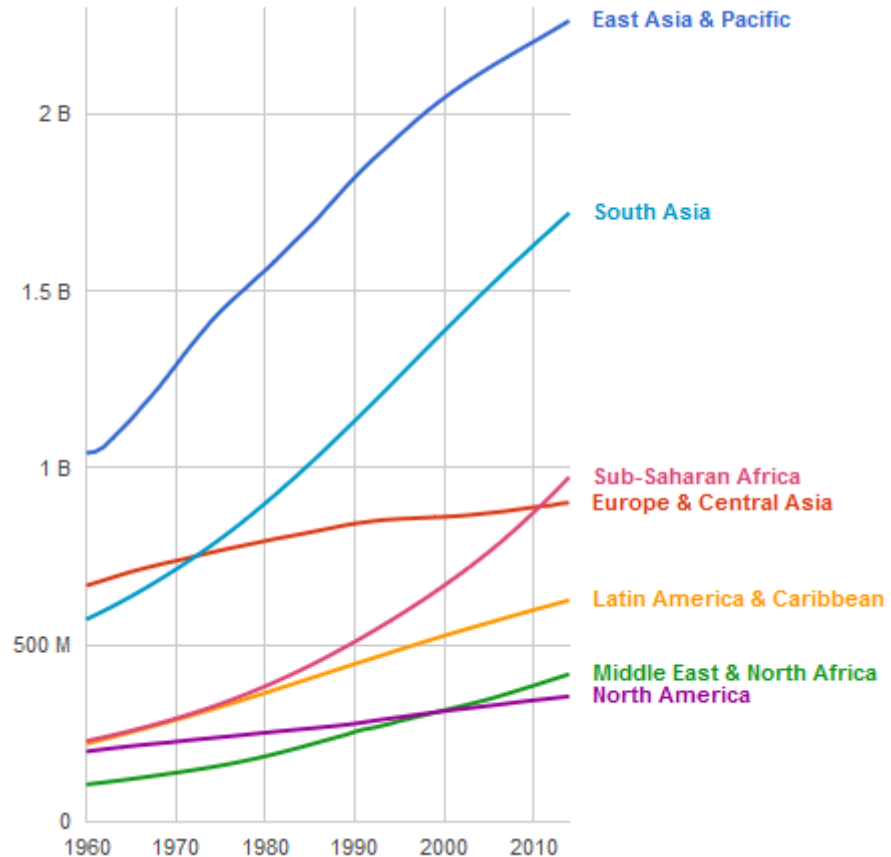
- Growing population
- Depleting natural resources
- Degrading environment
- Changing global climate
- Threatened ecosystem

Engineers have a key role to play in solving these

Population Growth

Impact of population growth on:

- Deforestation
- Food security
- Water & Waste
- Urbanization
- Poverty



Author: GregKaye
Population growth by World Bank continental division
https://en.wikipedia.org/wiki/World_population, FAL,
16 July 2016

Population Growth and Deforestation

Factors driving deforestation

- Agriculture
- Shifting cultivation
- Farming
- Fuel wood
- Logging
- Infrastructure

Deforestation leads to:

- Increasing atmospheric CO₂ burden and hence the warming of climate
- Ecological vulnerability and instability
- Resource scarcity
- Biological impoverishment
- Soil damage
- Hydrological problems (groundwater quality)

Population Growth and Food Security

“Population, when unchecked, goes on doubling every 24 years or increases in the geometric ratio,” while, food production or more generally, the means of subsistence - only increases at an arithmetic rate. (*Thomas Robert Malthus (1798), An Essay on the Principle of Population*)

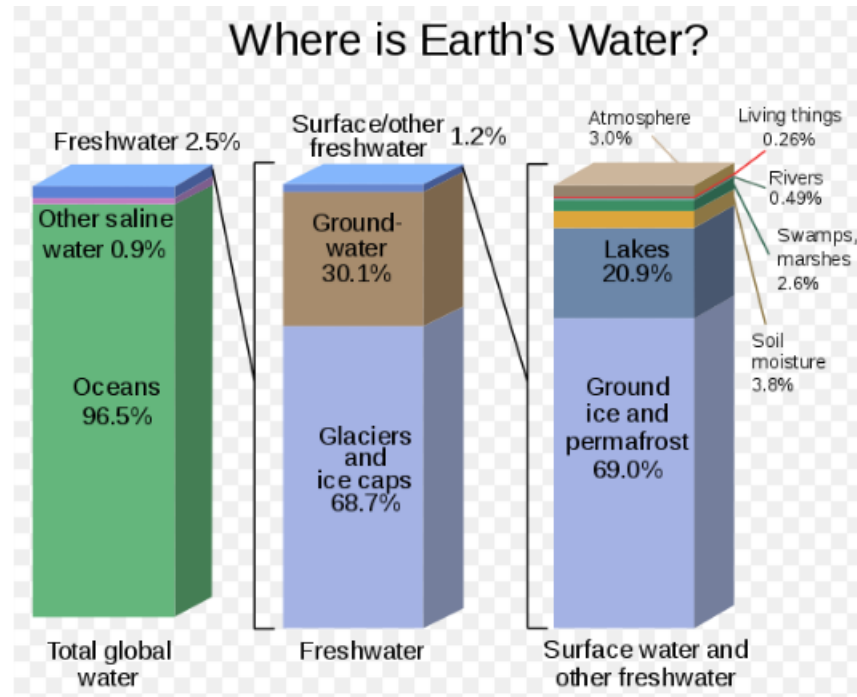
The Malthusian Dilemma

According to the World Bank and the United Nations, 1 to 2 billion humans are now malnourished, indicating a combination of insufficient food, low incomes, and inadequate distribution of food.

How can we provide more food for the world, with less land, less water and less fertilizer usage?

Population Growth and Water Resources

- The total volume of water on earth is about 1.4 billion km³
- The volume of freshwater resources is about 2.5% of the total volume
- 69% freshwater resources is in the form of ice and snow cover in mountains, the polar regions
- Around 30% freshwater is stored underground in the form of groundwater.
- Around 1% is surface and other freshwater of which approx 21% is in lakes and rivers



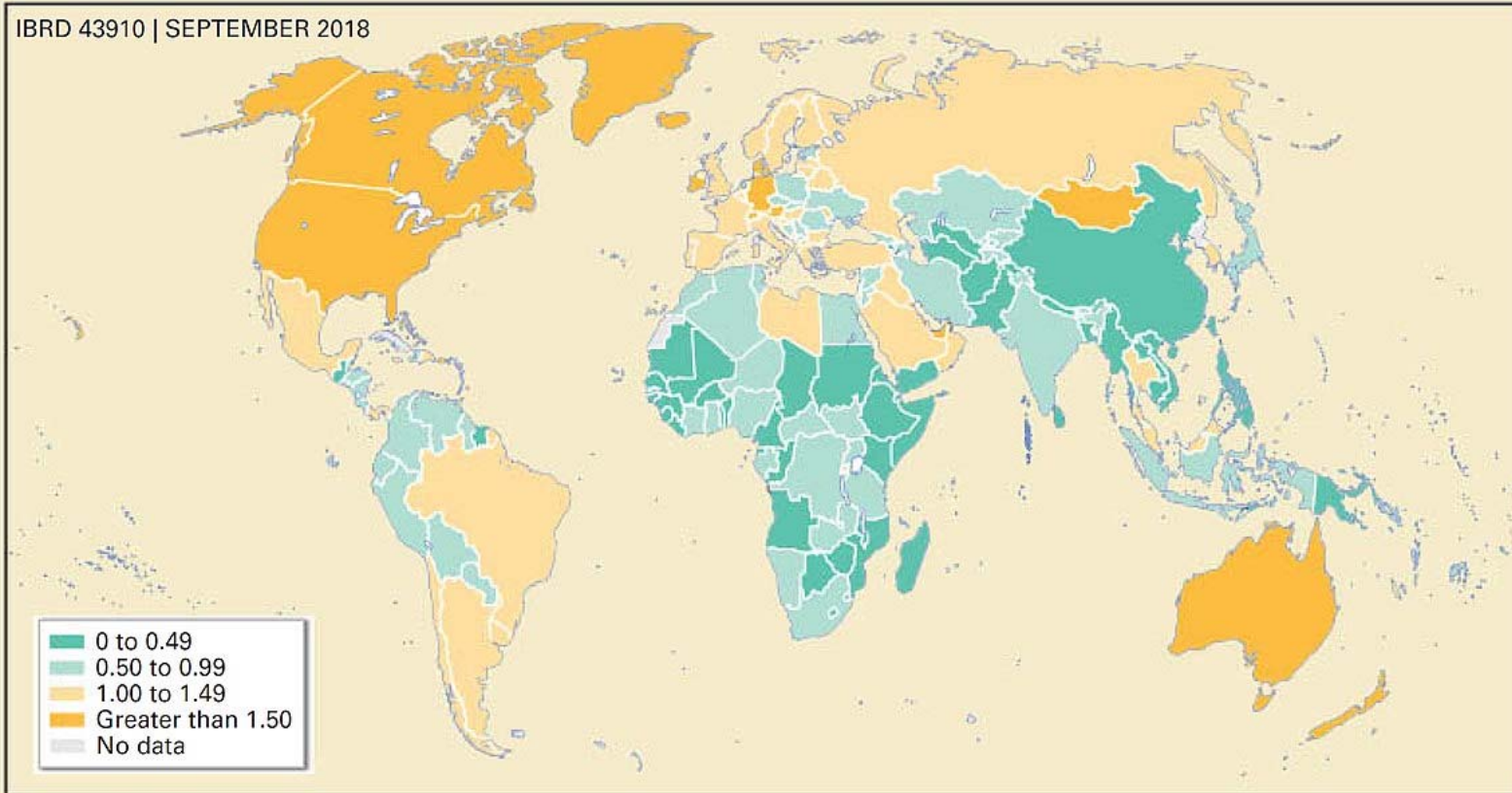
USGS, https://en.wikipedia.org/wiki/Water_resources

The world is not running out of water but it is not always available when and where people need it. Climate, normal seasonal variations, droughts and floods can all contribute to local extreme conditions.

Population and Waste

Map 2.1 Waste Generation Per Capita

IBRD 43910 | SEPTEMBER 2018



Note: kg = kilogram.

Source: Author Kaza, Silpa; Yao, Lisa C.; Bhada-Tata, Perinaz; Van Woerden, Frank.
https://commons.wikimedia.org/wiki/File:Waste_generation_per_day_per_capita_September_2018.jpg,
CC BY 3.0 IGO

Population and Waste - Composition of Waste Produced by Waikato

On average one person produces 528-591 kg waste per year

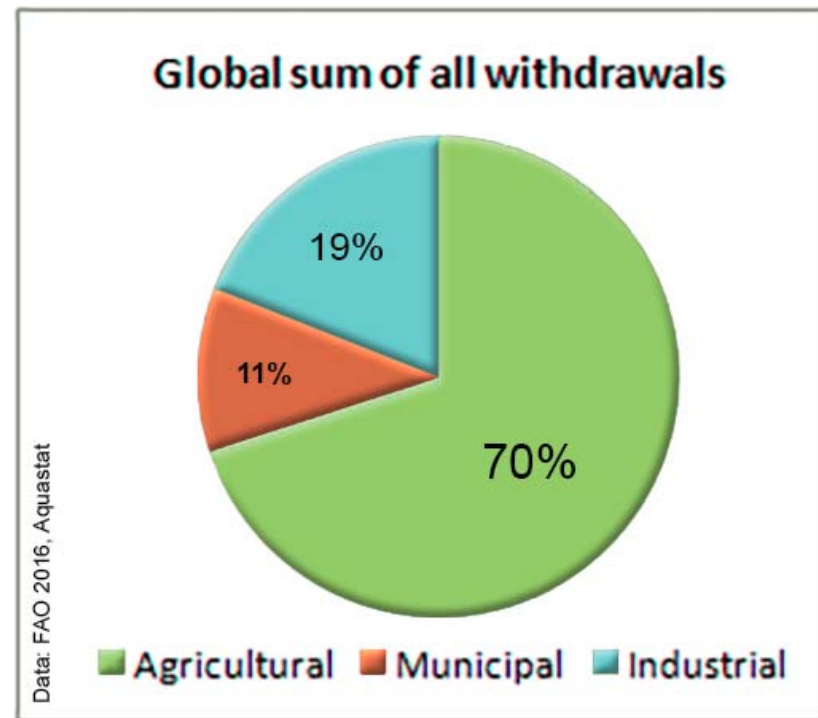
- kitchen scraps and garden clippings, 40%
- construction and demolition waste, 22%
- paper, 18%
- plastic, glass, metal
- potentially hazardous substances
 - Sanitary material, needles, bandages, etc



Credit: Muu-karhu
https://commons.wikimedia.org/wiki/File:Biodegradable_waste.jpg
CC BY-SA 3.0

Population Growth and Urbanisation

- 93% of the urbanization occurs in low income or developing countries
- Every second, the urban population grows by 2 people
- Each person needs 20-50 litres of water a day to ensure their basic needs (drinking, cooking and cleaning)
- More than one in six people worldwide - 894 million - don't have access to improved water sources
- By 2025, 1800 million people will be living in countries or regions with absolute water scarcity



Credit: Sampa,
https://en.wikipedia.org/wiki/Water_scarcity
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Population Growth And Poverty

There can be a negative relationship between rural poverty and environmental degradation due to:

- Low income communities depending heavily on local natural resources (forests, pastures, water) for their sustenance and often over-exploitation through low-productivity practices.
- Limited access to capital – low income communities are often unable to invest in resource conserving technologies.
- Lack of knowledge of sustainable resource management practices.

ENGEN370 Environmental Impact Assessment (Two)

Environmental Degradation

- Environmental degradation is defined as the deterioration of the environment through depletion of resources, destruction of ecosystems and the extinction of wildlife

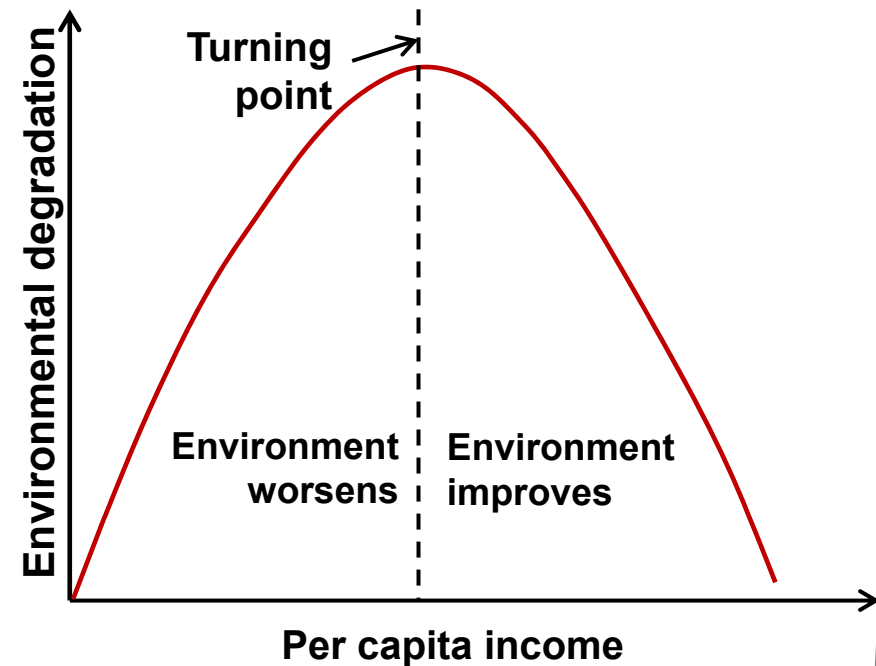
Environmental Degradation

Environmental degradation affects the health and well-being of peoples in a number of ways.

- Health problems due to release of contaminants into air, water, soil from industry and resource development projects.
- Availability of traditional food sources and medicines as a result of industrial contamination and the disruption of habitat.
- Impact on mental health and traditional ways of life.

The Kuznets Curve

- The environmental Kuznets* curve is a hypothesized relationship between various indicators of environmental degradation and income per capita.
- The pollution will first increase with the level of per capita income, reach maximum and then decrease at higher levels of income.



*Simon Kuznets-Russian American economist at Harvard University, won the 1971 Nobel Memorial Prize in Economic Sciences

The Kuznets Curve

Gradual decline of environmental degradation at high per capita income (after a certain level of economic growth) is due to:

- Greater value placed on the environment by affluent/progressive communities. Measures taken to affect this such as:
 - Movement towards information based industries and services, coupled with increased corporate environmental awareness.
 - Creation and enforcement of stricter environmental regulations.
 - Better/more carbon and energy efficient technologies and processes (less polluting industries)
 - Greater environmental expenditure (more investment in pollution abatement equipment and technology)

Global Warming

- Persistent human activities produce greenhouse gases (CO_2 , CH_4 and N_2O)
- These gases gather in the atmosphere, absorb and hold heat
- The more greenhouse gases we emit, the faster the world's climate heats up
- Carbon footprint is a measure of the exclusive total amount of CO_2 emissions that is directly and indirectly caused by an activity or is accumulated over the life stages of a product.
- Expressed as kilograms or tonnes CO_2 per person or activity

Global Warming

Commuting to work in an old petrol car:

Should you buy new (carbon cost of production)?

Should you keep the old till it falls apart (but less efficient)?

Buy second hand?

Petrol? Diesel? Hybrid? PHEV? Electric?

Work from home more?

Depends on a number of factors but studies have shown that around 75% of the life time emissions are from fuel consumed, not from production so keeping an old car isn't necessarily greener.

Buy second hand electric car perhaps (zero fuel, 80% renewable electricity in NZ)?

Roof (house) solar power

- 5 hours sunlight per day, 8 x 200W panels = 8 kWh
- 40 kWh electric car range is approximately 240 km
- 8kWh = 48 km (average NZ commute approximately 25km return)
- Ignoring various factors (e.g. panel to daytime storage, storage to car over night, weather etc)

Panels on car roof are under development.

Electric motorcycle option.

Input/Output Rule of Sustainability

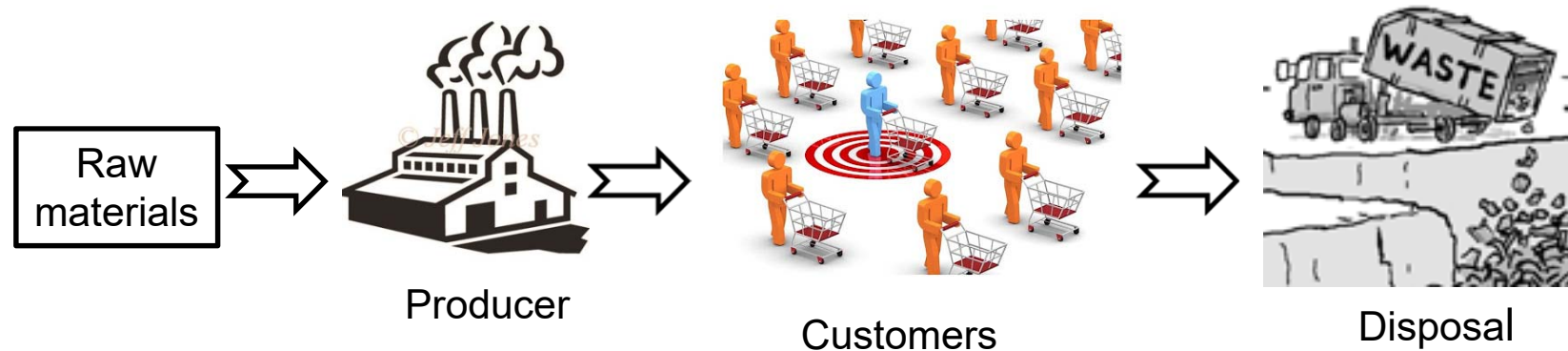
Input/Output Rule

- (a) Keep wastes within assimilative capacities.
- (b) Harvest within re-generative capacities of renewable resources.
- (c) Deplete non-renewables at the rate at which renewable substitutes are developed.

Carbon Footprint & Climate Neutral Production

- Reduce energy demand by implementing energy-saving measures.
- Use renewable sources of energy instead of fossil fuels.
- Produce and use energy with the most efficient practices that are available.
- Climate neutral production means producing with a net zero carbon footprint, or having net zero carbon emissions. This can be achieved by not emitting carbon at all or by compensation.
- Compensation can be done by balancing a measured amount of carbon emitted with an equivalent amount of abstracted or offset (e.g., by fixing CO₂ by planting trees)

Sustainable Development



Traditional supply chain model



Supply chain model influenced by sustainable development

Challenges and Why we need the EIA process

Summary

1. Maintaining essential ecological processes and life-support systems is essential.
2. Preserve genetic diversity and ensure sustainable utilization of species and ecosystems.
3. Use natural resources efficiently and effectively with minimal impact on environment.

EIA process is a key component of this.

ENGEN370 Environmental Impact Assessment (Four)

Environmental Impact Assessment

The EIA Process can include some or all of the following steps:

- Screening
- Scoping
- Alternatives
- Description of the project
- Description of the environment
- Identification of the impacts
- Effect of those impacts (prediction)
- Evaluation/assessment of significance of the impacts
- Mitigation measures to reduce impact
- Public consultation/participation
- Presentation (reporting)
- Review
- Decision making
- Monitoring
- Auditing

Environmental Impact Assessment

Screening

- Screening narrows the application of EIA to those projects that may have significant environmental impacts
- i.e. is an EIA actually needed for the proposal?

Consider 2 projects – is an EIA needed?

- Waikato expressway versus
- Installing solar panels on my roof

Environmental Impact Assessment

Scoping

- Scoping aims to identify at an early stage in the project or proposal those potential impacts that are deemed significant and should be included in the assessment

Environmental Impact Assessment

Alternatives

- Including alternatives to the proposed project ensures that the proposer has considered other:
 - Locations
 - Designs
 - Processes
 - Layouts
 - Operations
 - And the “Do Nothing” option

Environmental Impact Assessment

Description

- Project description – purpose of the project and its' stages, location and processes
- Environment description – states the present and future state of the environment within which the project is to be undertaken

Environmental Impact Assessment

Identifying Impacts

- Based on the previous steps, identifies the potentially significant impacts both +ve and –ve that may result from the project/proposal

Environmental Impact Assessment

Prediction and Evaluation of Impacts

- Aims to determine the magnitude of the impacts over the short, medium and long term on the environment versus the “Do Nothing” option
- Enables focus on the main adverse impacts

Environmental Impact Assessment

Mitigation

Identifies steps that can be taken to:

- Avoid
- Reduce
- Remedy
- Compensate

any significant adverse effects

Environmental Impact Assessment

Public consultation and participation

- Ensures the public are informed and have the opportunity to influence the decision making process

Presentation and Review

- Ensures the results are communicated effectively to the stakeholders (both technical and non-technical)
- Review of the EIA processes is essential for quality assurance

Environmental Impact Assessment

Decision making, monitoring and auditing

- Comes after the EIA is submitted, decision by the regulatory authority
- Monitoring is undertaken once/if decision is made to proceed to record the outcomes
- Auditing compares the actual outcomes versus the stated/predicted outcomes and provides valuable lessons learned for future EIA

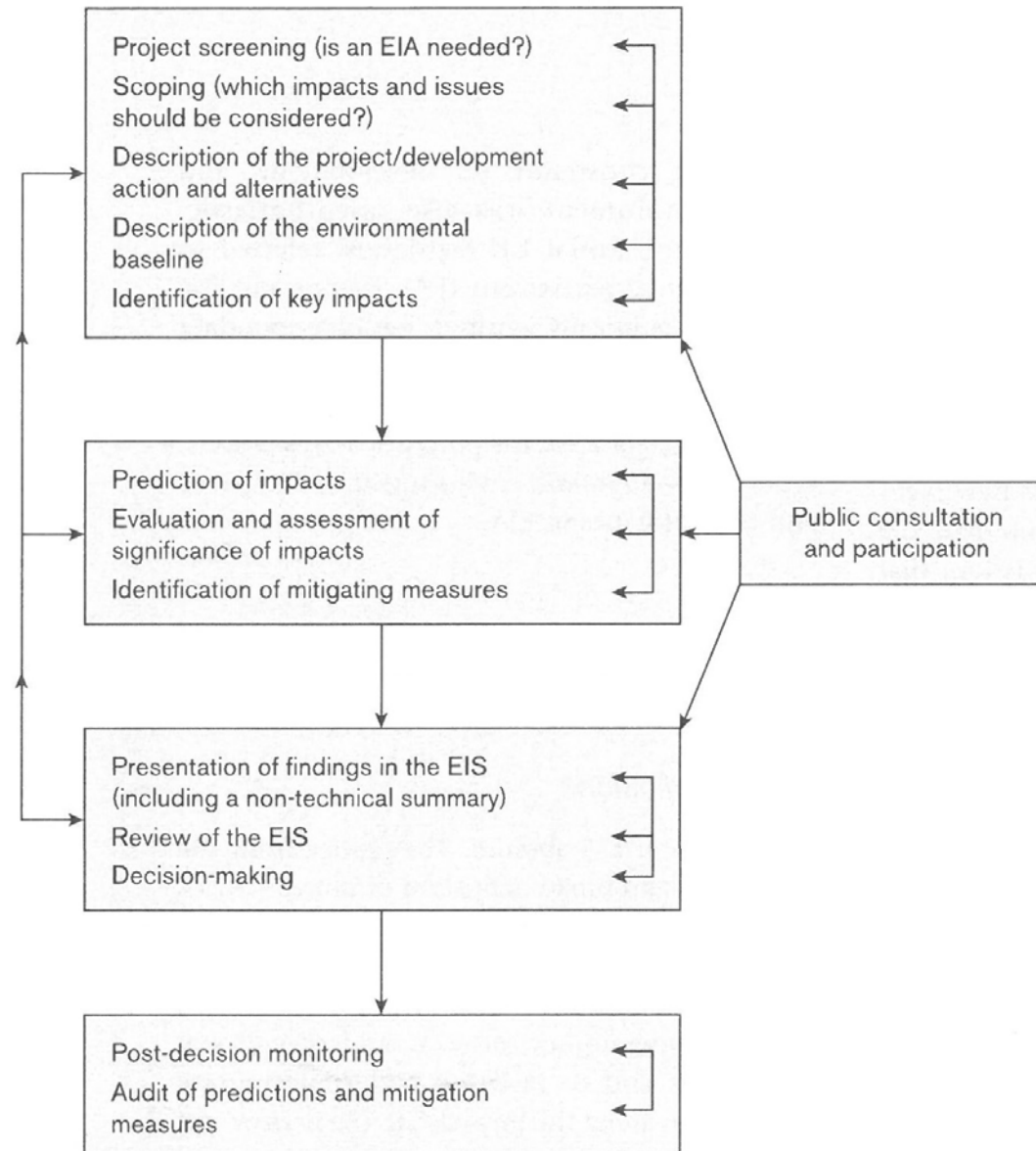


Figure 1.1

Important steps in the EIA process

Guide to Preparing an EIA

DOC has a simple step by step guide which can provide a basis or starting point for preparing your EIA for your project. See the reading list.

Step 1 Describe the environment

Step 2 Scope out the potential effects of your project

Step 3 Identify effects of your project and activities

Step 4 Describe any measures you can take to eliminate, avoid, remedy or mitigate the identified impacts

Step 5 Identify and evaluate alternatives

Step 6 Outline any monitoring programme you have developed

Plus consider risk and resilience

ENGEN370 Environmental Impact Assessment (Three)

Environmental Impact Assessment

- Purpose of an EIA
- Definition of an EIA
- The EIA process

Environmental Impact Assessment

Purpose of an EIA

- EIA is an aid to the decision making process
- Systematic examination of the environmental implications of a proposal and its' alternatives before making a decision to proceed
- It shouldn't be seen as yet another hurdle to be overcome
- It should be viewed as a useful tool for developers seeking to make the “right” decision
- An aid to the creation of actions early in the project life cycle to improve sustainability

Environmental Impact Assessment

Purpose of an EIA

- Projects can and will have a significant impact on stakeholders
- EIA serves as a tool that engages with stakeholders early and throughout the project life cycle
- Promotes consultation and participation by interested parties
- An instrument for sustainable development

The Pursuit of Growth and Negative Feedback

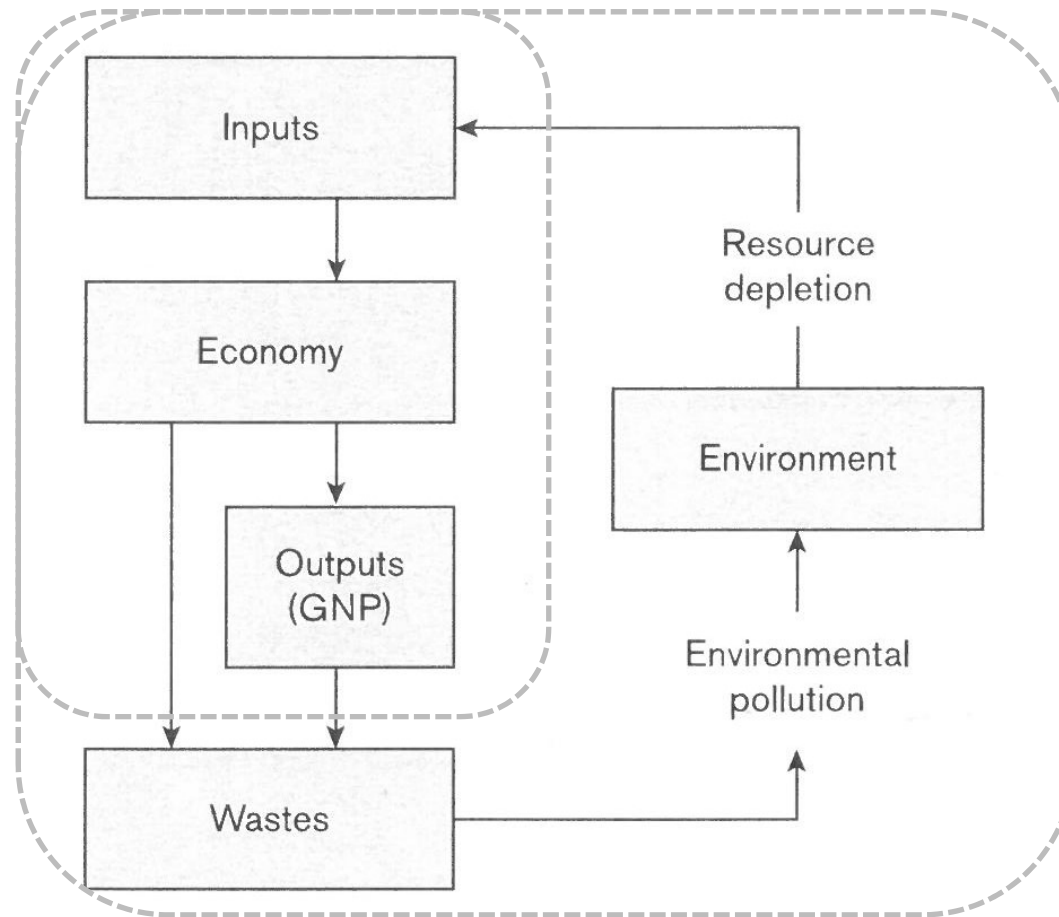
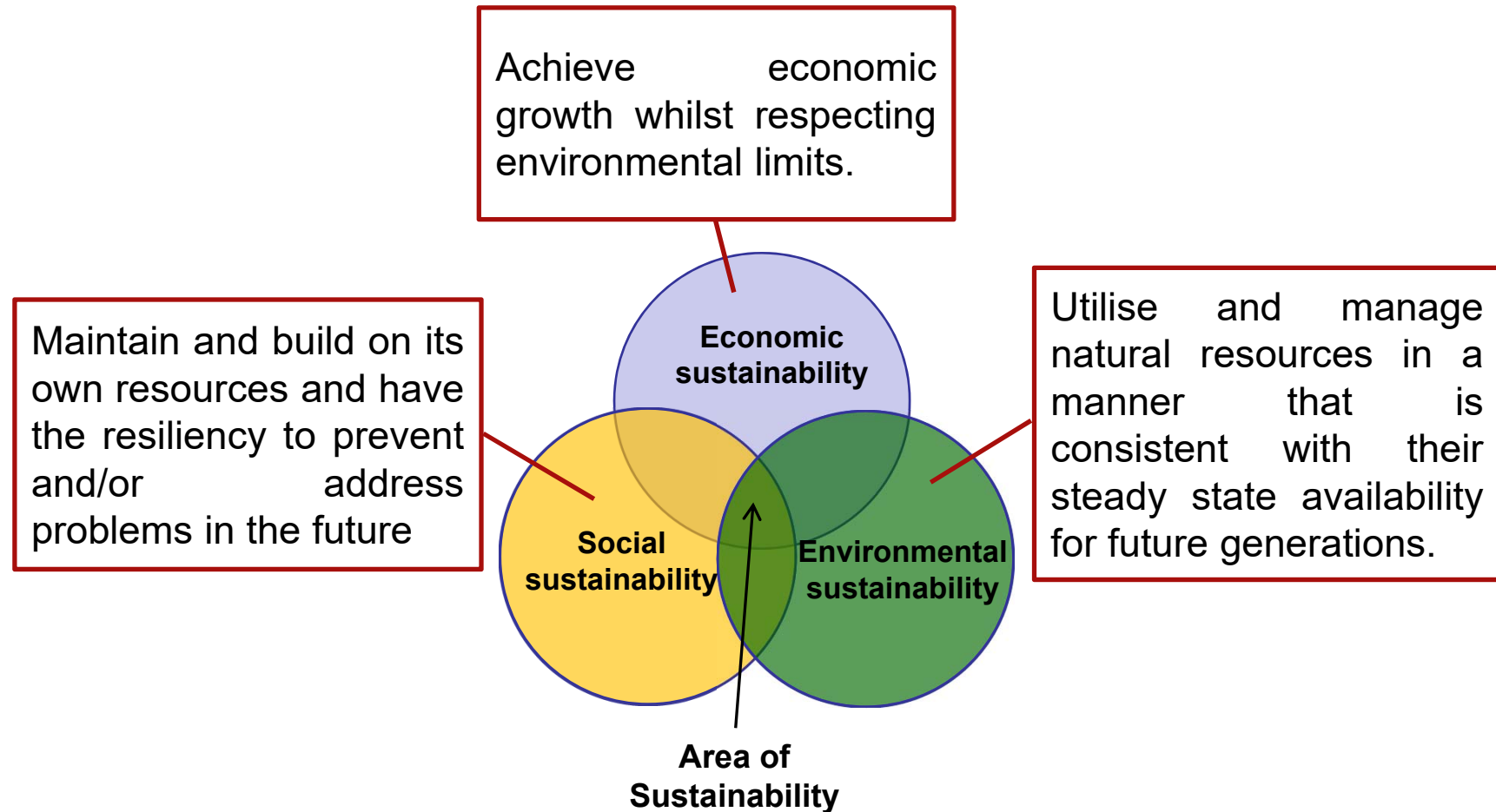


Figure 1.2

The economic development process in its environmental context (adapted from Boulding 1966)

Sustainable development models



Sustainable Development Models

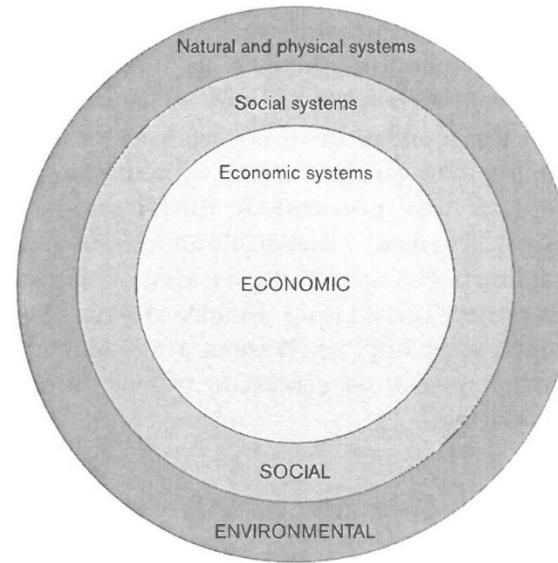


Figure 1.4

An alternative (hierarchical) perspective on the dimensions of sustainability

The environment is the foundation within which society operates.

Without these, can an economic system operate sustainably?

Objectives of Sustainable Development

Development (and hence tools such as EIA) should focus on:

- Efficient, effective use of natural resources
- Waste reduction and pollution prevention
- Energy efficiency
- New technologies and technology transfer
- Integrated environmental management system
- Environmental limits definition (regulations)
- Improving education, perception, attitude and behavioural changes
- Promoting social and cultural change

Environmental Impact Assessment

Definitions of what is an EIA – there are many. Some useful/not so useful. Still evolving

An early definition of EIA

- “To identify and predict the impact on the environment and on man’s health and well-being of legislative proposals, policies, programmes, projects and operational procedures, and to interpret and communicate information about the impacts” (Munn 1979)

Environmental Impact Assessment

Definitions of what is an EIA

This is better IMO:

- “The term environmental assessment describes a technique and a process by which information about the environmental effects of a project is collected, both by the developer and from other sources and taken into account by the planning authority in forming their judgements on whether the development should go ahead” – UK DoE (1989)

Environmental Impact Assessment

Definitions of what is an EIA

What about this one:

“An assessment of the impact of a planned activity on the environment” – UNECE (1991)

Helpful?

Environmental Impact Assessment

A more recent and more useful definition IMO that actually starts to break it down into steps i.e. what do you do:

- “the process of identifying, predicting, evaluating and mitigating the biophysical, social and other relevant effects of proposed development proposals prior to major decisions being taken and commitments made” – IAIA (2009)

It's a *process* comprising a series of *steps*

Environmental Impact Assessment

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